

Real world Scale Tor Traffic Correlation in Real Time

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Update

- ANN-aware Deep Metric Learning Loss
- CISC-S review & QnA session

Update

- ANN-aware Deep Metric Learning Loss

매 epoch 마다 k-nn 을 실행하여 LID 를 줄이는 방향으로 훈련하였을 때,

-> LID 를 줄이는 것은 성공, 20-dim->10-dim

-> 그러나 correlation 성능 향상으로 이루어지진 않음

Update

- CISC-S review & QnA session

-> Strength: GTT 데이터셋을 이용해 트래픽 규모 정량적 제시, 실행시간 단축 & 탐지 성능 개선

-> Weakness: 기존 DeepCoFFEA 를 확장한 엔지니어링적 기여

Update

- 그런데 사실 n^2 comparison 이 병목이라서 ANN 으로 $n \log n$ 으로 바꾸면 된다는 사실 자체는 꽤 obvious 한 것이라고 볼 수 있는 것 같다.
- Tor traffic correlation 자체에서는 한 사례가 없긴 하지만, 그것만으로는 약하다.
- 오히려, 왜 $n \log n$ 으로 줄여야만 하는가? 왜 그게 필요한가? 라는 질문이 더 흥미로운 것 같다.

Strength

- Tor traffic 의 real world scale 을 정량적으로 제시한 점에 더 집중해보기.

Related Work

- Real world scale 이라는 관점에서 related work 다시 정리
- 왜 이전 work 에서는 real world scale 을 고려하지 못했을까?
- Tor metric site only provide bandwidth/ GTT dataset (2024)

TABLE I
COMPARISON OF PRIOR TRAFFIC CORRELATION WORKS BY THREAT
MODEL AND EVALUATION SCALE.

Work	Threat Model	Eval. Scale
Feamster & Dingledine [1]	AS-level	Analytical
RAPTOR [6]	AS-level	50 clients, 50 websites
CTA [4]	AS-level	~50,000 flows
DeepCorr [3]	AS-level	~2,000 flows
DeepCoFFEA [5]	AS-level	~10,000 flows

Threat Model

- Both end side 를 보려면, attacker 보는 AS 내에 both Tor trace, Exit trace 가 지나가야 한다.
- Measuring and mitigating AS-level adversaries against Tor (Nithyanand et al.) 이라는 연구에서, 상위 10개국, 상위 200 사이트 기준 AS 가 겹칠 확률이 40퍼센트 가량 된다고 계산함.

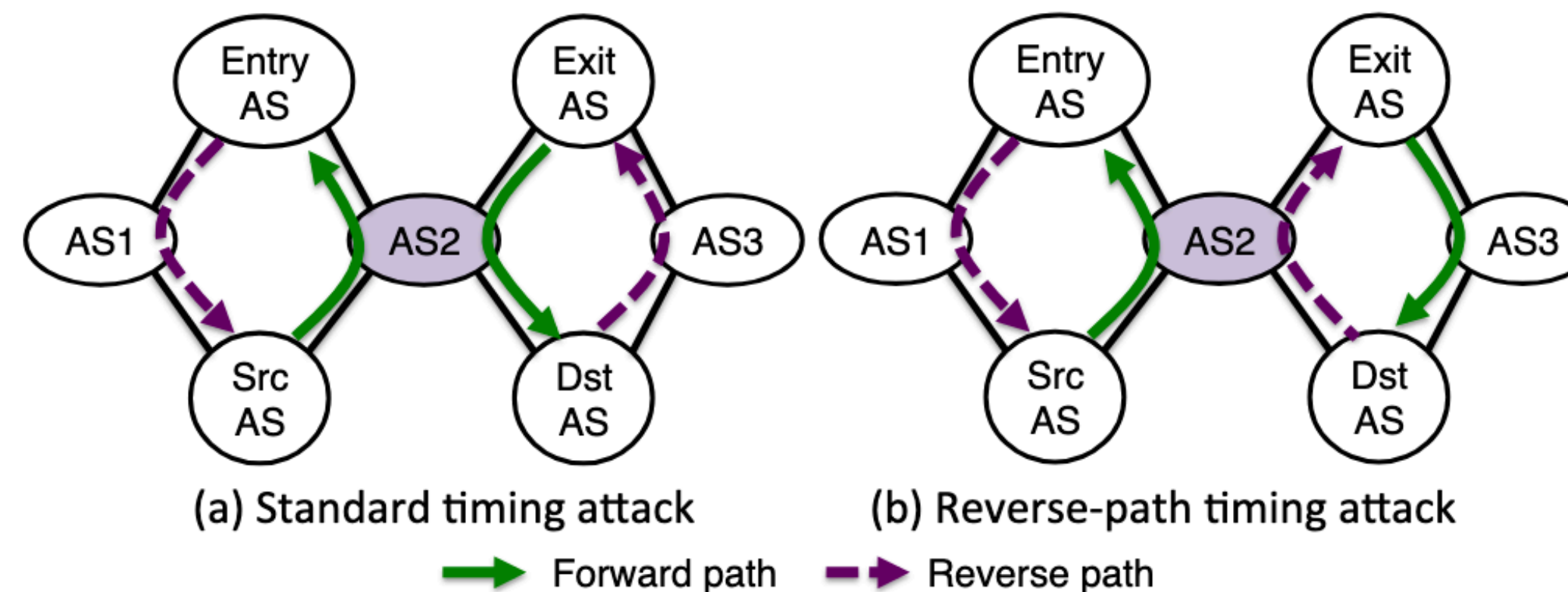


Fig. 1: Standard and reverse-path traffic correlation attacks. In the standard traffic correlation attack, AS2 must observe the direction of the connection that data is flowing on (forward path). In the reverse-path traffic correlation attack AS2 can infer the data flow using ACK numbers on the reverse path.

Measurement Study

- GTT 데이터셋을 이용한 역산
- 한 릴레이당 25k flow for 10 minute (Tor circuit lifecycle)
- 한 릴레이당 25Mib/s

TABLE II
CUMULATIVE EFFECT OF FILTERING STEPS ON FLOW VOLUME
(BUSY-WEEK SCENARIO)

Step	Retention	Criterion
Probabilistic sampling	2%	Busy-week circuits
Port-based filtering	96%	Ports 80 & 443 only
Connection validity	56%	$\geq 1,000$ packets
Per-relay normalization	1/8	Uniform relay assumption
Result	$\approx 24,000$ flows / relay / 10 min	

Measurement Study

- Tor metric website 에서 공개한 AS 단위의 스펙
- Tor 는 주기적으로 릴레이들의 bandwidth 를 측정하여 consensus 문서를 배포함

Country	Autonomous System	Consensus Weight	Advertised Bandwidth	Guard Probability	Middle Probability	Exit Probability	Relays	Guard	Exit
(18 distinct)	OVH SAS (AS16276)	11.5724%	16998.75 MiB/s	19.5066%	14.8182%	0.3922%	608	555	15
 Netherlands	Church of Cyberology (AS215125)	11.4433%	15529.77 MiB/s	3.6870%	4.7127%	25.9308%	630	126	476
(2 distinct)	Hetzner Online GmbH (AS24940)	7.4017%	9710.59 MiB/s	11.8065%	10.3984%	0.0000%	358	255	0
 Germany	Stiftung Erneuerbare Freiheit (AS60729)	6.3873%	8711.37 MiB/s	1.1520%	2.8803%	15.1300%	319	97	258
 United States	1st Amendment Encrypted Openness LLC (AS36849)	5.7006%	12500.12 MiB/s	9.8500%	7.2516%	0.0000%	1004	987	0
 Austria	Foundation for Applied Privacy (AS208323)	4.1952%	5034.71 MiB/s	2.4718%	2.5071%	7.6071%	153	29	120
(5 distinct)	IONOS SE (AS8560)	3.8182%	5375.67 MiB/s	5.4484%	5.8628%	0.1432%	273	192	2
 Germany	netcup GmbH (AS197540)	3.7417%	5230.66 MiB/s	4.6315%	4.7495%	1.8440%	299	238	57
 Sweden	QuxLabs AB (AS214503)	3.1105%	4017.64 MiB/s	0.9520%	0.7585%	7.6215%	156	35	124
(2 distinct)	FranTech Solutions (AS53667)	1.9942%	2844.53 MiB/s	0.7682%	0.6996%	4.5149%	172	131	133
(3 distinct)	Scaleway SAS (AS12876)	1.3639%	1894.16 MiB/s	2.1632%	1.7666%	0.1618%	82	73	7
 Sweden	Foreningen for digitala fri- och rattigheter (AS198093)	1.3285%	2478.8 MiB/s	0.2714%	0.2714%	3.4429%	149	148	148

<https://metrics.torproject.org/rs.html#aggregate/as>

Measurement Study

- GTT 데이터셋이 측정되었을 때의 bandwidth 과 각 AS 의 bandwidth 를 계산에 포함

TABLE III

ESTIMATED 10-MINUTE FLOW VOLUME PER AS BASED ON EXIT RELAY BANDWIDTH, ORDERED BY CONSENSUS WEIGHT (DESCENDING), SOURCED FROM TOR METRICS (MAY 2026). FLOW COUNT IS EXTRAPOLATED FROM THE GTT23 DATASET (25 MiB/s $\approx$ 25,000 FLOWS/10 MIN). ASes WITH NO EXIT RELAYS ARE OMITTED.

Autonomous System	Consensus Weight (%)	Exit Relays	Avg BW (MiB/s)	Flows/relay/10 min	Total flows/10 min
OVH SAS (AS16276)	11.5724	15	28.0	27,958	419,370
Church of Cyberology (AS215125)	11.4433	476	24.7	24,650	11,733,400
Stiftung Erneuerbare Freiheit (AS60729)	6.3873	258	27.3	27,308	7,045,464
Foundation for Applied Privacy (AS208323)	4.1952	120	32.9	32,907	3,948,840
IONOS SE (AS8560)	3.8182	2	19.7	19,691	39,382
netcup GmbH (AS197540)	3.7417	57	17.5	17,494	997,158
QuxLabs AB (AS214503)	3.1105	124	25.8	25,754	3,193,496
FranTech Solutions (AS53667)	1.9942	133	16.5	16,538	2,199,554
Scaleway SAS (AS12876)	1.3639	7	23.1	23,100	161,700
Foreningen for digitala fri- og rettigheter (AS198093)	1.3285	148	16.6	16,636	2,462,128
Gigahost AS (AS56655)	1.2937	52	24.0	23,983	1,247,116
1337 Services GmbH (AS210558)	1.2655	73	23.1	23,104	1,686,592
F3 Netze e.V. (AS205100)	1.1064	21	59.9	59,876	1,257,396
KeFF Networks Ltd (AS41281)	0.8478	37	31.2	31,246	1,156,102
Florian Kolb (AS58087)	0.7301	18	38.3	38,276	688,968
SABOTAGE LLC (AS61125)	0.6937	17	44.2	44,249	752,233
FlokiNET ehf (AS200651)	0.6190	44	14.8	14,782	650,408
Fourplex Telecom LLC (AS27284)	0.6129	74	15.6	15,555	1,151,070
MOJI SAS (AS213873)	0.5993	16	40.9	40,941	655,056
The Infrastructure Group B.V. (AS60404)	0.5853	33	15.4	15,441	509,553
Init7 (Switzerland) Ltd. (AS13030)	0.4904	2	24.1	24,074	48,148

Experiment

- DeepCoFFEA 의 경우, 150k 일 때 이미 10분을 넘김.
- Proximity Graph 를 이용한 방법의 경우, 200k 이여도 아직 10초 수준.

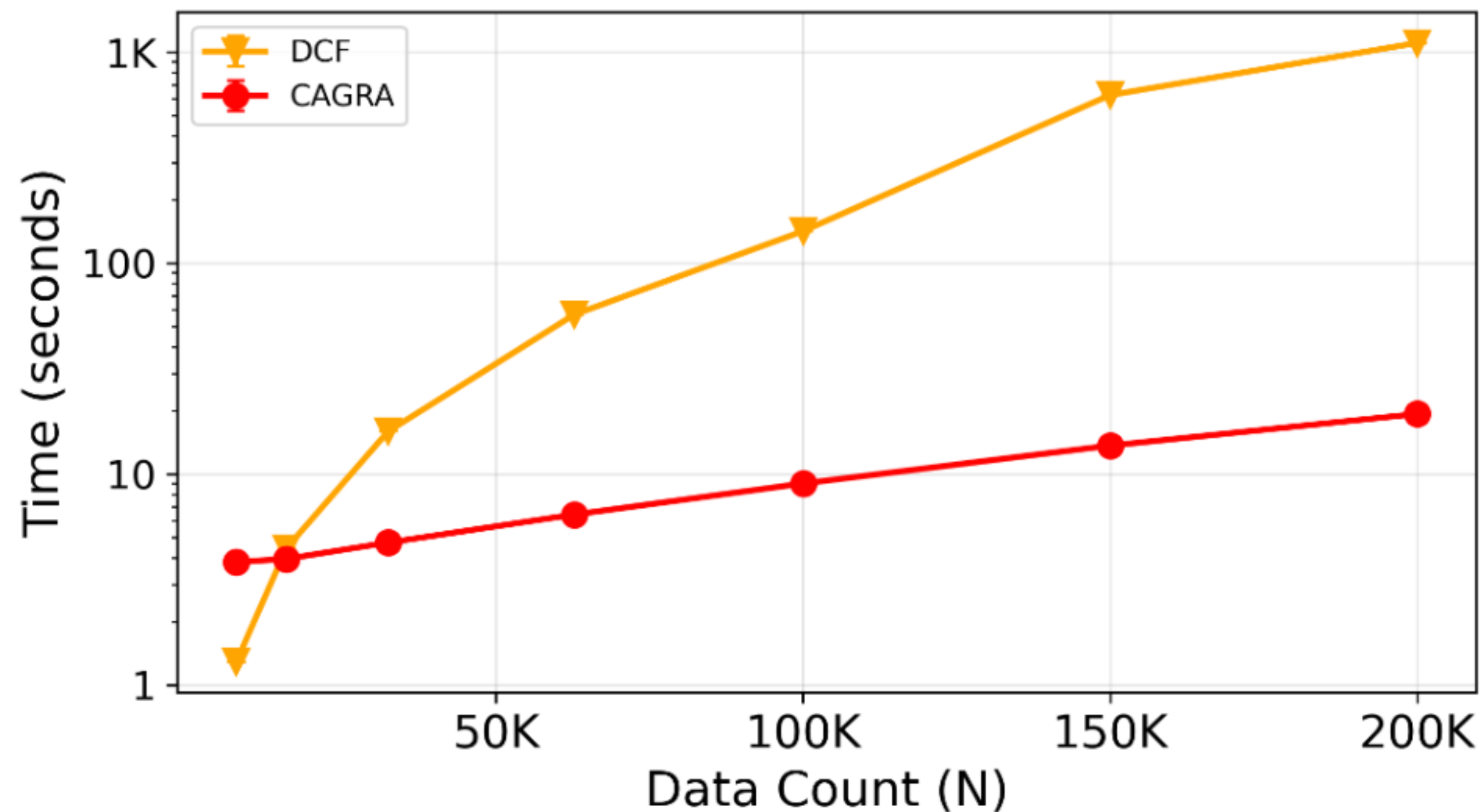


Fig. 2 Running Time Comparison

Future Plan

- Experiment with real world scale of AS level attacker
- Ablation study with ANNs